CMSC433 / Part 02

Functor, Applicative, and State types

Topic mock

Topic mock, using the sample's question styles. Attempt independently; feedback after the whole file. Untimed practice, no points assigned. Current exam scope and rules are unconfirmed.

Give the most general type or "ill-typed" with a reason. All parts are independent. Keep function-valued payloads and necessary constraints.

(a) `get >>= put`

Answer: [Answer.]

(b) `modify tail`

Answer: [Answer.]

(c) `put 'x' >> put True`

Answer: [Answer.]

(d) `(:)  <$> get`

Answer: [Stored type; returned type; whole type.]

(e) `\f mx -> f <$> mx`

Answer: [Most general type.]

(f) `get >>= \x -> return (x, x + 2)`

Answer: [Most general type and constraint.]

Supplied signatures

```
get :: State s s
put :: s -> State s ()
modify :: (s -> s) -> State s ()
tail :: [a] -> [a]
(:) :: a -> [a] -> [a]
(<$>) :: Functor f => (a -> b) -> f a -> f b
(>>=) :: Monad m => m a -> (a -> m b) -> m b
(>>) :: Monad m => m a -> m b -> m b
return :: Monad m => a -> m a
```